INCH-POUND

AN775 Rev 15
28 March 2018
SUPERSEDING
AN775 Rev 14
w/AMENDMENT 1
12 June 2012

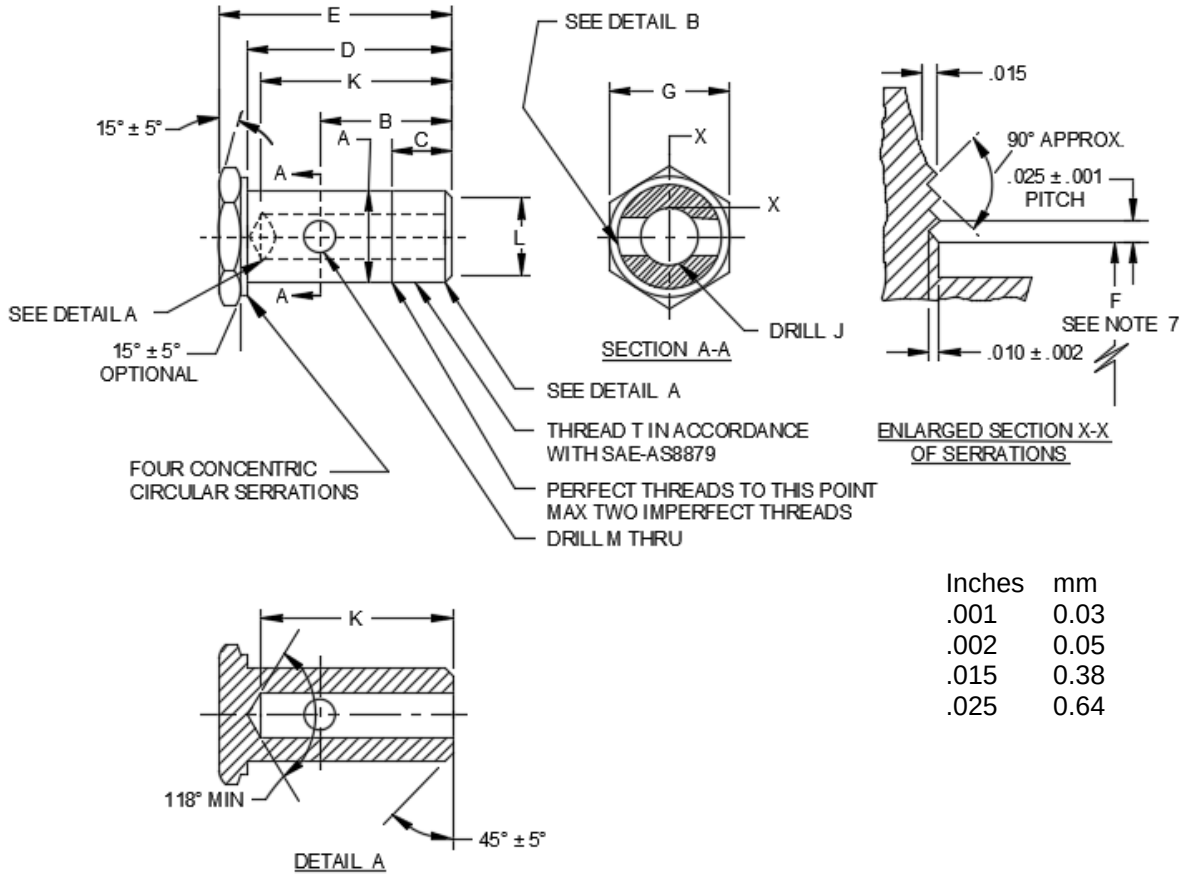
DETAIL SPECIFICATION SHEET

BOLT - FLUID PASSAGE

Reinstated after 19 July 2011. Inactive for new design.
For new design, use SAE-AS5161.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4875.



Inches	mm
.001	0.03
.002	0.05
.015	0.38
.025	0.64

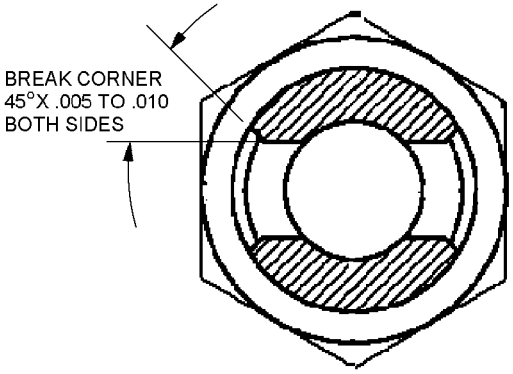
See notes at end of figure.

FIGURE 1. Bolt - fluid passage dimensions and configurations.

AMSC N/A

FSC 4730

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:16Z
Check the source to verify that this is the current version before use.



Inches	mm
.005	0.13
.010	0.25

DETAIL B

Dash number	Tubing OD inches (mm)	Thread T (Ref) SAE-AS8879	A Dia. +.000 -.004 (0.10) inches (mm)	B ±.010 (0.25) inches (mm)	C inches (mm)
-4	.250 (6.35)	.4375-20UNJF-3A	.435 (11.05)	.781 (19.84)	.438 (11.13)
-5	.313 (7.95)	.5000-20UNJF-3A	.498 (12.65)	.781 (19.84)	.438 (11.13)
-6	.375 (9.53)	.5625-18UNJF-3A	.560 (14.22)	.875 (22.23)	.468 (11.89)
-8	.500 (12.70)	.750-16UNJF-3A	.748 (19.00)	1.094 (27.79)	.562 (14.27)
-10	.625 (15.88)	.8750-14UNJF-3A	.873 (22.17)	1.250 (31.75)	.594 (15.09)
-12	.750 (19.05)	1.0625-12UNJ-3A	1.060 (26.92)	1.468 (37.29)	.656 (16.66)
-16	1.000 (25.40)	1.3125-12UNJ-3A	1.310 (33.27)	1.687 (42.85)	.656 (16.66)

Dash number	D inches (mm)	E inches (mm)	F Dia. +.010 (0.25) -.000 (see note 7) inches (mm)	G Hex (ref) (see note 5) inches (mm)	
-4	1.188 (47.96)	1.375 (34.93)	.469 (11.91)	.688 (17.48)	±.004 (0.10)
-5	1.188 (47.96)	1.375 (34.93)	.531 (13.49)	.750 (19.05)	±.004 (0.10)
-6	1.344(34.14)	1.562 (39.67)	.594 (15.09)	.813 (20.65)	±.004 (0.10)
-8	1.688 (42.88)	1.938 (49.23)	.781 (19.84)	1.000 (25.40)	±.004 (0.10)
-10	1.969 (50.01)	2.250 (57.15)	.906 (23.01)	1.125 (28.58)	±.005 (0.13)
-12	2.312 (58.72)	2.625 (66.68)	1.094 (27.79)	1.375 (34.93)	±.005 (0.13)
-16	2.812 (71.42)	3.188 (80.98)	1.344 (34.14)	1.625 (41.28)	±.016 (0.41)

See notes at end of figure.

FIGURE 1. Bolt - fluid passage dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:16Z
Check the source to verify that this is the current version before use.

Dash number	J Dia. inches (mm)	K inches (mm)	L Dia. inches (mm)	M Dia. inches (mm)
-4	.172 (4.37)	1.156 (29.36)	.344 (8.74)	.188 (4.78)
-5	.234 (5.94)	1.156 (29.36)	.406 (10.31)	.219 (5.56)
-6	.297 (7.54)	1.312 (33.32)	.469 (11.91)	.250 (6.35)
-8	.391 (9.93)	1.656 (42.06)	.625 (15.88)	.313 (7.95)
-10	.484 (12.29)	1.938 (49.23)	.750 (19.05)	.375 (9.53)
-12	.609 (15.47)	2.156 (54.76)	.938 (23.83)	.438 (11.13)
-16	.844 (21.44)	2.578 (65.48)	1.188 (30.18)	.625 (15.88)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Break sharp edges .and remove all hanging burrs and slivers.
4. Unless otherwise noted tolerances shall be ±.005 inches (0.13 mm) and angles ±.5°.
5 Tolerances for dimensions G and N are in accordance with SAE-AS4395 style E.
6. Machined surfaces shall be smooth to 125µin Ra, hex surfaces may be 250µin Ra, unless otherwise specified on the figures. Surface finish shall be in accordance with ASME B46.1.
7. Diameter F is last pitch diameter of last serration from point to point.
8. Dimensions A, F, and J shall be concentric within .005 inch (0.13 mm) total indicator movement.
9. For design features purposes, this standard takes precedence over documents referenced herein.

FIGURE 1. Bolt - fluid passage dimensions and configurations - Continued.

REQUIREMENTS:

Dimensions and configuration shall be in accordance with figure 1.

Size -2 through -16 aluminum alloy maximum operating pressure is 1500 psi (10 MPa).

Steel, corrosion resistant steel (CRES), and titanium maximum operating pressure is 3000 psi (21 MPa).

Except as otherwise noted materials shall be in accordance with SAE-AS4875, see table I for material code.

TABLE I. Material and code letters.

Code letter	Material
F	Steel alloy 4130 or 4120 In accordance with SAE-AMS6370 or SAE-AMS6382 1/
J	CRES, type 304
K	CRES, type 316
R	CRES, type 321
T	Titanium alloy 2/
W	Aluminum alloy 7075-T73
WN	Aluminum alloy 7075-T73 anodize in accordance with SAE-AS4875 then apply NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A

1/ Steel alloy 125,000 psi (860 MPa) minimum tensile strength.
2/ Not for use in oxygen systems.

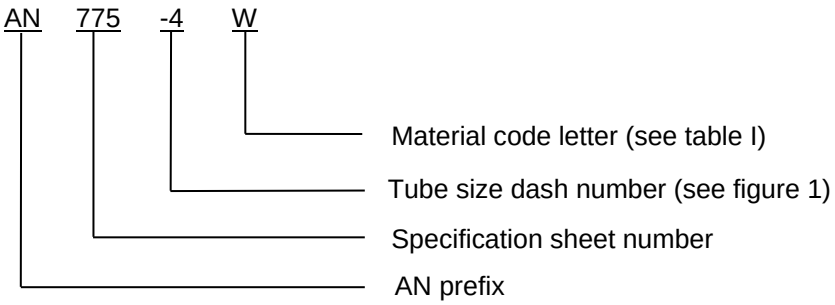
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:16Z
Check the source to verify that this is the current version before use.

Unless otherwise specified finishes shall be in accordance with SAE-AS4875.

Trivalent wrenchability. Applies to aluminum fittings that receive the NAVAIR TCP coating in accordance with MIL-DTL-81706, type II. When the finish has been damaged due to poor wrenchability, the surface of the fluid passage bolt shall be touched up using the brush plating process below. The term "trivalent wrenchability" is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching.

Brush plating of NAVAIR TCP shall be in accordance with MIL-DTL-81706, type 2, class 1A, material form 1 through 6, application method B. Example of a PIN: M817062A6B.

Part or Identifying Number (PIN): The PIN consists of the letters “AN” the specification sheet number, a dash number for tube size, and a material code letter for material type. Unassigned PIN’s shall not be used.



PIN example: AN775-4W indicates a bolt - fluid passage, .250 inches (6.35 mm) tube OD, .4375-20UNJF-3A (7/16) threaded end, aluminum alloy 7075-T73.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Cadmium is prohibited from use in NAVSEA owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2.

Class I and II ozone depleting substances (ODS) shall not be used in AN775 or any referenced procedures.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Supersession data:

Steel blank designator has been replaced by steel alloy “F” designator.

Due to stress corrosion cracking aluminum alloys 2014 and 2024 “D” designator has been replaced by aluminum alloy 7075 “W” designator example: AN774-D4 use AN774-W4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:16Z
Check the source to verify that this is the current version before use.

Table II provides a detailed cross-reference of cancelled AN775 PINs and replacement SAE-AS5161 PINs. Users are cautioned to evaluate replacements for their particular application.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

TABLE II. Cross-reference data.

AN PIN (inactive part)	Tube size	Replacement AS PIN (new design part)	Replacement AN PIN (inactive part)
AN775-4	.250	AS5161-04	AN775-4F
AN775-4D	.250	AS5161W04	AN775-4W
AN775-4J	.250	AS5161J04	
AN775-4K	.250	AS5161K04	
AN775-4R	.250	AS5161R04	
AN775-4T	.250	AS5161T04	
AN775-4W	.250	AS5161W04	
AN775-4WN		Not available	
AN775-5	.313	AS5161-05	AN775-5F
AN775-5D	.313	AS5161W05	AN775-5W
AN775-5J	.313	AS5161J05	
AN775-5K	.313	AS5161K05	
AN775-5R	.313	AS5161R05	
AN775-5T	.313	AS5161T05	
AN775-5W	.313	AS5161W05	
AN775-5WN		Not available	
AN775-6	.375	AS5161-06	AN775-6F
AN775-6D	.375	AS5161W06	AN775-6W
AN775-6J	.375	AS5161J06	
AN775-6K	.375	AS5161K06	
AN775-6R	.375	AS5161R06	
AN775-6T	.375	AS5161T06	
AN775-6W	.375	AS5161W06	
AN775-6WN		Not available	
AN775-8	.500	AS5161-08	AN775-8F
AN775-8D	.500	AS5161W08	AN775-8W
AN775-8J	.500	AS5161J08	
AN775-8K	.500	AS5161K08	
AN775-8R	.500	AS5161R08	
AN775-8T	.500	AS5161T08	
AN775-8W	.500	AS5161W08	
AN775-8WN		Not available	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:16Z
Check the source to verify that this is the current version before use.

TABLE II. Cross-reference data - Continued.

AN PIN (inactive part)	Tube size	Replacement AS PIN (new design part)	Replacement AN PIN (inactive part)
AN775-10	.625	AS5161-16	AN775-10F
AN775-10D	. 625	AS5161W16	AN775-10W
AN775-10J	. 625	AS5161J16	
AN775-10K	. 625	AS5161K16	
AN775-10R	. 625	AS5161R16	
AN775-10T	. 625	AS5161T16	
AN775-10W	. 625	AS5161W16	
AN775-10WN		Not available	
AN775-12	.750	AS5161-12	AN775-12F
AN775-12D	.750	AS5161W12	AN775-12W
AN775-12J	.750	AS5161J12	
AN775-12K	.750	AS5161K12	
AN775-12R	.750	AS5161R12	
AN775-12T	.750	AS5161T12	
AN775-12W	.750	AS5161W12	
AN775-12WN		Not available	
AN775-16	1.00	AS5161-16	AN775-16F
AN775-16D	1.00	AS5161W16	AN775-16W
AN775-16J	1.00	AS5161J16	
AN775-16K	1.00	AS5161K16	
AN775-16R	1.00	AS5161R16	
AN775-16T	1.00	AS5161T16	
AN775-16W	1.00	AS5161W16	
AN775-16WN		Not available	

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to SAE-AS4875, this document references the following:

MIL-DTL-81706
ASME B46.1
SAE-AMS6370
SAE-AMS6382
SAE-AS4395
SAE-AS5161
SAE-AS8879
T9070-AL-DPC-020/077-2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:16Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2018-043)

Review activities:
Navy - SH
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:16Z
Check the source to verify that this is the current version before use.